

Strategy Extension Memo: Capital, Hedge, ATRP Dynamic Sizing, Parallel Testing

Updated through full empirical loop with operator-driven synthesis of fixes

Part 1: Capital Sizing, Unhedged Baseline

Question: At the current strategy calibration (0.0005 BTC lots, approximately \$50 in USD at prevailing prices, ~1 satoshi minimum profit delta), how much capital is needed to survive 6-week trading windows without forced cuts?

Method: 364 days of historical BTC daily prices, 322 rolling 6-week windows. Calibrated from actual Dec 2024 to Jan 2025 run: ~0.018 BTC of bag growth per 1% drawdown, ~\$1,782 of capital tied up per 1% of drawdown at the BTC price levels observed during the run. Capital required for a given window = (max drawdown in pct) x (\$1,782 per 1%).

Result:

Coverage target	Capital required	Drawdown survived
50% of windows (median)	~\$18,000	up to -12%
90% of windows	~\$48,000	up to -29%
95% of windows	~\$51,000	up to -35%
99% of windows	~\$56,000	up to -35%

Tail caveat: dataset is recent 365 days only and excludes 2018, 2022, and Mar 2020 BTC drawdowns. A more honest distribution including those tails pushes the 95th percentile to roughly \$87k unhedged.

The original \$21,971 of deployed capital covered about 53% of historical 6-week windows. Forced cuts at that capital level were the median outcome at the calibration tested, not bad luck.

Part 2: Capital Sizing, With Perp Hedge

Question: how does the same capital sizing change when a perp futures hedge is added at hedge ratio h , leverage L ?

Method: same rolling-window analysis as Part 1. Capital requirement formula adjusted to:

$$\text{capital_required} = \text{bag_cost_basis} \times (1 + h/L - h \times d_frac)$$

where $(1 + h/L)$ accounts for spot capital plus perp margin requirement, and $(- h \times d_frac)$ accounts for the perp short's PnL credit during the drawdown that can be reallocated to spot. With $L = 10$, the formula becomes $\text{capital} = \text{bag} \times (1.1 - h \times d_frac)$. Deeper drawdowns make the hedge more capital-efficient.

Result:

Coverage	Unhedged	h = 0.5	h = 0.75	h = 1.0
50th (median)	\$18,061	\$18,098	\$18,116	\$18,134
90th	\$46,836	\$42,889	\$40,630	\$38,270
95th	\$49,775	\$44,639	\$42,119	\$39,809
99th	\$54,034	\$47,224	\$43,819	\$40,839

The hedge's value is concentrated in tail drawdowns. Median-case capital is essentially unchanged. The 95th percentile is reduced by 20% at $h=1.0$; tail-adjusted reduction reaches roughly 40% at deep drawdowns.

Live backtest finding (separate study): the hedge has a real running cost in non-drawdown periods. Replaying the actual Nov 2024 to Jan 2025 spot tape with synthetic perp hedge applied: about \$1,300 cost per 80 days at $h=1.0$, or roughly \$6-8k per year of drag in benign markets. The hedge wins on multi-year horizons where drawdowns are inevitable; it loses on benign single-year horizons. Recommended ratio in practice: **$h=0.5$** captures most of the tail benefit at half the directional drag.

Part 3: The Fix Framework

Six fixes identified from the post-mortem. Four are active improvements. Two are demoted or subsumed.

Part 3a: Fix #1 — Capital Sized to the Drawdown Distribution

The binding constraint in the original run was capital, not strategy logic. The round-trip engine worked. The capacity-relief mechanism fired because capital ran out before the drawdown reversed.

From the rolling-window analysis in Parts 1 and 2: about \$30k with $h=0.5$ hedge or \$50k unhedged covers 95% of recent 6-week windows. Tail-adjusted including 2018, 2022, and Mar 2020 pushes that to \$100k unhedged. The original \$21,971 covered about 53% of historical windows. Forced cuts were the median outcome, not bad luck.

This is the primary fix. Everything else is secondary.

Part 3b: Fix #2 — Dynamic Lot Sizing for Fee Tier Maintenance

Objective: keep 30-day rolling notional above the \$10M free-tier threshold regardless of daily fill rate variation.

Signal: fill count, not price movement. The lot size has nothing to do with how BTC is moving.

Formula: $\text{lot} = \text{clip}(\text{daily_notional_target} / \text{fills_per_day}, \$72, \$200)$

At the target fill rate of 281 fills per hour, the daily notional target of \$333k implies a base lot of \$72 - the free tier floor. On quiet days with fewer fills the lot rises to compensate. On busy days it drops toward minimum. The tier is maintained continuously rather than by accident.

Key distinction: lot size is a revenue-neutral parameter for tier qualification when considered alone. However lot size and delta interact through profit economics: revenue per round trip = lot_btc x delta. If delta is held fixed while lot size changes, the revenue-to-lot-size ratio drifts. Fix #3 resolves this by targeting profit per round trip (P*) directly, so delta adjusts automatically when lot size changes. The two fixes are designed to work together.

Part 3c: Fix #3 — Profit-Target-Driven Dynamic Delta

Delta - how far above the buy price the paired sell sits - is not the primary control variable. It is a derived output. The primary control variable is P*, target profit per round trip in dollars.

The relationship: $\text{delta} = P^* / \text{lot_btc}$

This matters because lot size changes with fill count (Fix #2) to maintain the free tier. If delta were held fixed while lot size changes, the revenue-to-lot-size ratio would drift - when lot rises in low-vol, a fixed delta means each fill earns proportionally less per dollar of capital deployed. Targeting P* instead preserves profitability across lot size changes automatically. When lot rises, delta compresses. When lot falls, delta expands.

Formula: $P^* = \text{clip}(P^*_{\text{base}} \times \text{ATRP}_{24\text{h}} / \text{ATRP}_{\text{median}}, P^*_{\text{floor}}, P^*_{\text{ceiling}})$

Delta is then derived each cycle: $\text{delta} = P^* / \text{lot_btc}$

P* parameter derivation: the floor and ceiling are expressed in delta space at the reference lot of \$72, then converted to P* space.

- Floor: delta \$10 (carry cost meets spread income) -> $P^*_{\text{floor}} = \$10 \times (\$72 / \$99,012) = \0.0073
- Ceiling: delta \$100 (24-hour fill rate drops below 95.8% on the actual tape) -> $P^*_{\text{ceiling}} = \$100 \times (\$72 / \$99,012) = \0.0727
- Geometric midpoint in delta space: $\sqrt{\$17.6 \times \$69.0} = \$34.9$ -> $P^*_{\text{base}} = \$34.9 \times (\$72 / \$99,012) = \0.0254

At $P^*_{\text{base}} = \$0.0254$ and a \$72 lot the implied delta is \$34.9 - the same geometric midpoint derived in the original design. The P* framing adds no new discretion. It simply expresses the same calibrated number in profit space rather than price space, and makes the lot-size interaction explicit.

Three guardrails defend against regime-transition risk:

1. P* floor at \$0.0073 (equivalent to delta \$10 at \$72 lot). Never gets unbounded-tight.
2. Asymmetric response speed. Faster ATR window (ATR(7)) for widening, slower (ATR(28)) for tightening. Reacts quickly to expanding range, slowly to contracting.
3. Hard P* ceiling at \$0.0727 (equivalent to delta \$100 at \$72 lot). Derived from fill rate analysis.

Why P* rather than delta: professional market makers reason about spread in earnings-per-round-trip terms, not in price-distance terms. Delta as a primary input requires re-derivation whenever lot size changes. P* as a primary input is regime-invariant in its logic - the system targets a dollar amount of earnings per cycle and lets the price distance follow from position size and market conditions. One revenue control variable, one position sizing variable, one market condition signal.

Behavior By Regime

Regime	ATRP value	Sizing response	Protection mode
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Narrow range (below median)	Below median	P* compresses, delta tightens	Yield enhancement
Median range	At median	Base configuration (0.0005 BTC lot, P*=\$0.025, delta=\$35)	None
Wide range (above median)	Above median	P* expands, delta widens	Revenue optimization

Part 3d: Fix #4 — Soft Bag Cap as Defense-in-Depth

~~Hard bag cap as primary risk control~~ demoted. Replaced by a soft alarm threshold.

If cumulative bag exceeds 2x design-time peak inventory, the bot pauses and alerts the operator. No automatic market-sells. Human decides next action. Defense-in-depth against hedge failure - not the primary risk control.

Part 3e: Fix #5 — Demoted

~~Static smaller delta in benign regimes~~ subsumed by Fix #3. ATRP dynamic delta handles this continuously without a separate rule.

Part 3f: Fix #6 — Perp Futures Hedge at h=0.5

Sized to capture meaningful tail protection while limiting directional drag in benign markets.

From the backtest in Part 2: h=0.5 is the efficient point - about half of the capital benefit at half the directional drag. h=1.0 captures the full benefit but doubles the drag in benign markets. Running cost approximately \$663 over 80 days at h=0.5.

Operational rule: maintain perp short at roughly 0.5 x spot inventory. Rebalance when deviation exceeds 10% of target. Use limit orders on the perp side to minimize taker fees.

Optional outer layer: if BTC closes below the 200-day MA and drawdown from the 365-day high exceeds 25%, pause the bot entirely. Resume when both conditions reverse. This catches confirmed multi-month bears that ATRP alone cannot fully throttle.

Part 4: Multi-Bot Parallel Parameter Testing

Rather than backtest parameter sensitivity - subject to overfitting and look-ahead bias - run multiple instances with different parameters in parallel on live markets. Each instance gets isolated capital but shares operational infrastructure (account, fee tier qualification, logging).

Phase 1: 2x2 grid (base sensitivities)

Bot	Lot size	Delta
A (base)	\$50	1 satoshi

B	\$50	2 satoshi
C	\$100	1 satoshi
D	\$100	2 satoshi

Phase 2: Add Bot E with dynamic lot (fill count) and ATRP delta, normalized to the base configuration. Validate against the static grid.

Phase 3: Add Bot F with ATRP-dynamic sizing and $h=0.5$ hedge. Validate hedge contribution during drawdown events.

Practical constraints: all bots run from one account so notional aggregates to the \$10M tier. Each bot needs \$7-10k of isolated capital. At roughly 2,000 round trips per day per bot, performance differences become statistically meaningful within 2-3 weeks.

Part 5: Build Order

1. Implement ATRP(24h) computation and visualize regime against historical BTC data
2. Implement dynamic lot sizing (fill count signal) and ATRP-dynamic delta in the Python framework, with the three guardrails
3. Apply to the Dec 2 to Jan 21 trade tape and quantify cut reduction
4. Implement the soft bag cap alarm
5. Set up Phase 1 multi-bot infrastructure (one account, unified logging, isolated capital per bot)
6. Run Phase 1 for 30 days. Document results.
7. Add Bot E for Phase 2. Run another 30 days.
8. Add Bot F with hedge for Phase 3. Run another 30 days.

End state after roughly 3 months: a fully tested, capital-budgeted, dynamically-sized, optionally-hedged strategy with live performance attribution across parameter dimensions.

Closing Note on the Lagging-Indicator Concern

- **Fix #2 (dynamic lot)** uses fill count as the signal. Revenue-neutral - adjusts to maintain tier qualification, not to improve edge.
- **Fix #3 (ATRP delta)** uses range as a measurement unit, not a signal. ATR measures how much price moved, not where it is going. No prediction.
- **Optional outer-layer filter** uses lagging indicators only as posture descriptors for the rare regime where they are obviously correct. Posture is not prediction.
- **No fix attempts to predict price direction.** All defensive logic is either reactive (state-based) or structural (capital sizing, hedge).
- **No fix requires backtest-based parameter tuning.** All thresholds are round numbers from canonical references; all sizing is parametric in range.

The combination of range-aware sizing with three guardrails, an optional lagging posture filter, a hedge sized to half the spot bag, and parallel live testing for parameter validation is the closest possible construction to "a defensive system the operator can trust without overfitting."

Calibration note: numbers are derived from actual Dec 2, 2024 to Jan 21, 2025 Kraken trade history and 364 days of recent BTC daily prices. ATRP work outlined here has not yet been computed and is suggested as the next step.